

```

user@deb:~$ minikube start --driver=docker --cni=calico --cpus=4 --memory=4096
🐹 minikube v1.37.0 on Debian 13.1 (vbox/amd64)
🔧 Using the docker driver based on user configuration
🔧 Using Docker driver with root privileges
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚚 Pulling base image v0.0.48 ...
🔥 Creating docker container (CPUs=4, Memory=4096MB) ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔧 Configuring Calico (Container Networking Interface) ...
🔍 Verifying Kubernetes components...
   ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass

! /usr/bin/kubectl is version 1.32.3, which may have incompatibilities with Kubernetes 1.34.0.
   ■ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
user@deb:~$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

user@deb:~$ minikube kubectl version
> kubectl.sha256: 64 B / 64 B [-----] 100.00% ? p/s 0s
> kubectl: 57.75 MiB / 57.75 MiB [-----] 100.00% 26.83 MiB p/s 2.4s
Client Version: v1.34.0
Kustomize Version: v5.7.1
Server Version: v1.34.0
user@deb:~$ kubectl get nodes
NAME          STATUS    ROLES          AGE   VERSION
minikube      Ready     control-plane   34m   v1.34.0
user@deb:~$ kubectl get pods -A
NAMESPACE     NAME                                                    READY   STATUS    RESTARTS   AGE
kube-system   calico-kube-controllers-59556d9b4c-sk8vn              1/1     Running   1 (38m ago) 38m
kube-system   calico-node-65mgn                                       1/1     Running   0           38m
kube-system   coredns-66bc5c9577-mntjn                              1/1     Running   2 (37m ago) 38m
kube-system   etcd-minikube                                          1/1     Running   0           38m
kube-system   kube-apiserver-minikube                               1/1     Running   0           38m
kube-system   kube-controller-manager-minikube                     1/1     Running   0           38m
kube-system   kube-proxy-45zp8                                       1/1     Running   0           38m
kube-system   kube-scheduler-minikube                               1/1     Running   0           38m
kube-system   storage-provisioner                                    1/1     Running   1 (37m ago) 38m
user@deb:~$ █

```

```

user@deb:~/Documents/lab2$ ls
deployment.yaml  'lab2 report.odt'
user@deb:~/Documents/lab2$ kubectl apply -f deployment.yaml
deployment.apps/wikis created
user@deb:~/Documents/lab2$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
wikis-f5b699b6f-6ll4c              0/1     ContainerCreating   0           28s
wikis-f5b699b6f-6vt54              0/1     ContainerCreating   0           28s
wikis-f5b699b6f-blbw7              0/1     ContainerCreating   0           28s
user@deb:~/Documents/lab2$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
wikis-f5b699b6f-6ll4c              1/1     Running   0           102s
wikis-f5b699b6f-6vt54              1/1     Running   0           102s
wikis-f5b699b6f-blbw7              1/1     Running   0           102s
user@deb:~/Documents/lab2$ kubectl apply -f service.yaml
service/wiki-svc created
user@deb:~/Documents/lab2$ minikube service wiki-svc --url
http://192.168.49.2:30080

```

After accessing this URL and attempting to install wiki.js, an error occurs :

EACCES: permission denied, mkdir '/data'.

That's why the Wiki.js installation fails after entering the initial username and password.

Wiki.js needs to write configuration files to /data but /data either doesn't exist in the container or the container doesn't have write permission for it.

As a result, the installation stops right after the admin account creation.

We need a correct Kubernetes Yaml deployment by mounting a volume on /data to persist data (with write permissions for the container to /data) and to configure a PostgreSQL database for Wiki.js.

Correct Yaml deployment :

```
! deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: wikijs
5    labels:
6      app: wikijs
7  spec:
8    replicas: 3
9    selector:
10     matchLabels:
11       app: wikijs
12    template:
13     metadata:
14       labels:
15         app: wikijs
16     spec:
17       containers:
18         - name: wikijs-container
19           image: lscr.io/linuxserver/wikijs:latest
20           ports:
21             - containerPort: 3000
22           env:
23             - name: DB_TYPE
24               value: postgres
25             - name: DB_HOST
26               value: postgres
27             - name: DB_PORT
28               value: "5432"
29             - name: DB_USER
30               value: wikijs
31             - name: DB_PASS
32               value: wikijsrocks
33             - name: DB_NAME
34               value: wikijs
35           volumeMounts:
36             - mountPath: /data
37               name: wikijs-data
38       volumes:
39         - name: wikijs-data
40           emptyDir: {}
41
```

```
postgres.yaml
1  apiVersion: v1
2  kind: PersistentVolumeClaim
3  metadata:
4    name: postgres-pvc
5  spec:
6    accessModes:
7      - ReadWriteOnce
8    resources:
9      requests:
10       storage: 1Gi
11  ---
12  apiVersion: apps/v1
13  kind: Deployment
14  metadata:
15    name: postgres
16  spec:
17    selector:
18      matchLabels:
19        app: postgres
20    template:
21      metadata:
22        labels:
23          app: postgres
24      spec:
25        containers:
26          - name: postgres
27            image: postgres:15
28            env:
29              - name: POSTGRES_DB
30                value: wikijs
31              - name: POSTGRES_USER
32                value: wikijs
33              - name: POSTGRES_PASSWORD
34                value: wikijsrocks
35            ports:
36              - containerPort: 5432
37            volumeMounts:
38              - mountPath: /var/lib/postgresql/data
39                name: postgres-storage
40            volumes:
41              - name: postgres-storage
42                persistentVolumeClaim:
43                  claimName: postgres-pvc
44  ---
45  apiVersion: v1
46  kind: Service
47  metadata:
48    name: postgres
49  spec:
```

```
49 spec:
50   selector:
51     app: postgres
52   ports:
53     - port: 5432
54       targetPort: 5432
```

```
user@deb:~/Documents/lab2$ kubectl apply -f service.yaml
service/wiki-svc unchanged
user@deb:~/Documents/lab2$ kubectl apply -f postgres.yaml
persistentvolumeclaim/postgres-pvc created
deployment.apps/postgres created
service/postgres created
user@deb:~/Documents/lab2$ kubectl apply -f deployment.yaml
deployment.apps/wikijs configured
user@deb:~/Documents/lab2$ minikube service wiki-svc --url
http://192.168.49.2:30080
```

The nodeport service is now operational and allows access to Wiki.js from outside the cluster, /data is writable and Wiki.js can save user credentials.



Brought to you by linuxserver.io

GID/UID	
User UID:	911
User GID:	911
Linuxserver.io version: v2.5.308-ls197	
Build-date: 2025-10-24T13:44:38+00:00	

```
2025-10-29T14:46:25.731Z [MASTER] info:
2025-10-29T14:46:25.731Z [MASTER] info: ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲
Connection to localhost (127.0.0.1) 3000 port [tcp/*] succeeded!
[ls.io-init] done.
```

```
user@deb:~/Documents/lab2$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
postgres-5d45787877-cwmg2	1/1	Running	0	20m
wikijs-85c4ccd4cb-jj5mm	1/1	Running	0	19m
wikijs-85c4ccd4cb-p2jkg	1/1	Running	0	20m
wikijs-85c4ccd4cb-xmqdx	1/1	Running	0	19m

Wiki.js after installing the application with a new account :



Access to the administration page :

The screenshot shows a web browser window with the address bar displaying "Admin | Wiki.js" and the URL "http://192.168.49.2:30080/a/dashboard". The page is titled "Wiki.js" and "ADMINAREA". The left sidebar contains a navigation menu with the following items:

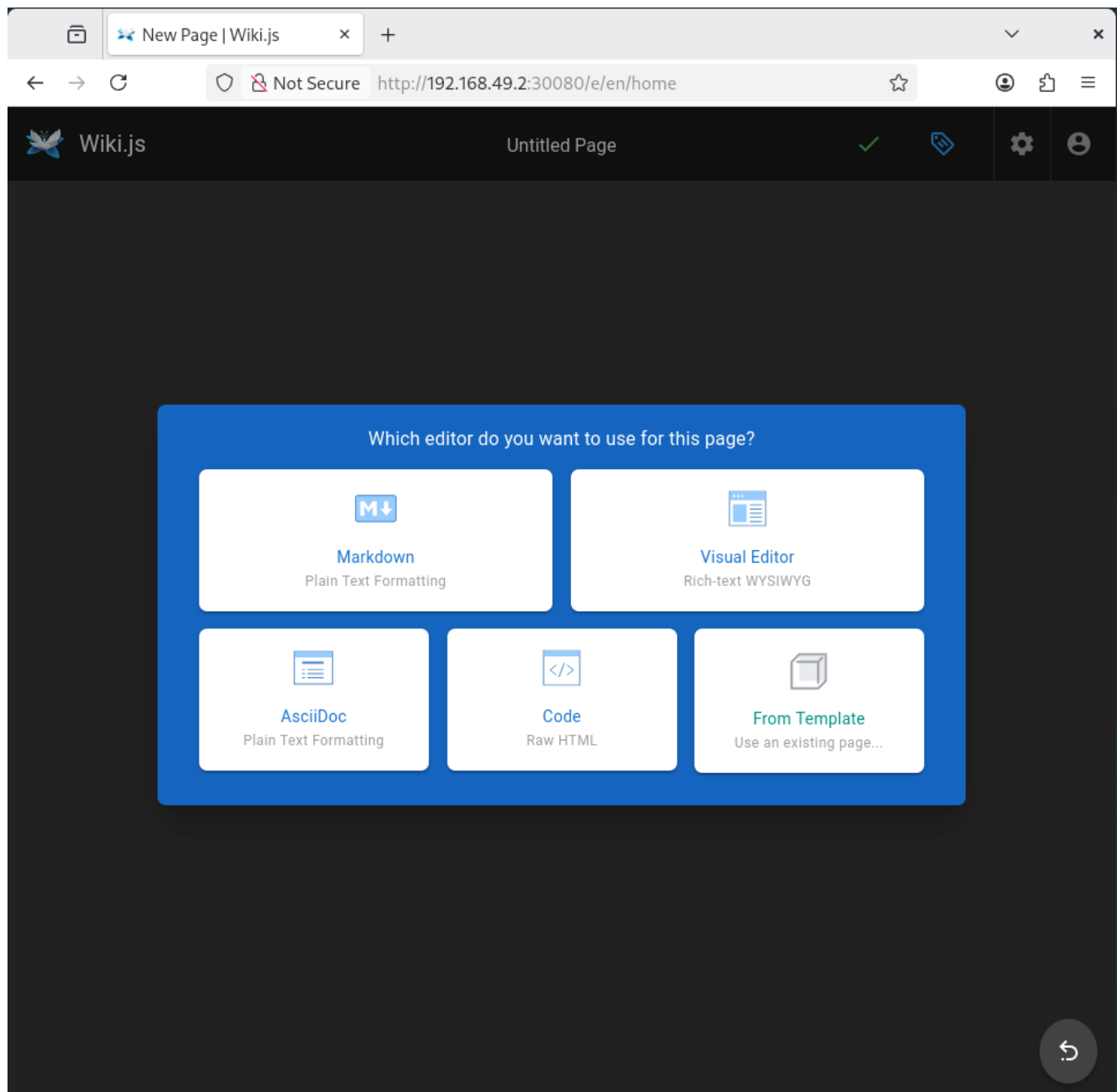
- nav.site
 - dashboard.title
- nav.users
 - groups.title
 - users.title
- nav.modules
 - analytics.title
 - auth.title
 - comments.title
 - rendering.title
 - search.title

The main content area displays the "dashboard.title" section with a subtitle "dashboard.subtitle". It features two blue cards: "DASHBOARD.PAGES" with a value of 0 and "DASHBOARD.GROUPS" with a value of 0. Below these are two white cards: "DASHBOARD.RECENTPAGES" and "DASHBOARD.LASTLOGINS", both showing "No data available". At the bottom is a light blue card titled "contribute.title" with a subtitle "dashboard.contributeSubtitle" and "dashboard.contributeHelp", and a button labeled "DASHBOARD.CONTRIBUTELEARNMORE".

On the right side, there is a user profile dropdown menu for "Administrator" (alecomte@lerebours.fr) with options for "Profile" and "Logout". Below this is a green banner with the text "Wiki.js n/a" and "dashboard.versionLatest".

The footer of the page states "Powered by Wiki.js".

Editor's selection for creating our home page :



```

user@deb:~/Documents/lab2$ kubectl apply -f test-pod.yaml
pod/test-container created
user@deb:~/Documents/lab2$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
postgres-5d45787877-cwmq2          1/1     Running   0           76m
test-container                      1/1     Running   0           37s
wikijs-85c4ccd4cb-jj5mm            1/1     Running   0           75m
wikijs-85c4ccd4cb-p2jkg            1/1     Running   0           76m
wikijs-85c4ccd4cb-xmqdx            1/1     Running   0           75m
user@deb:~/Documents/lab2$ kubectl exec -it test-container -- bash

```

```

root@test-container:/# nslookup wiki-svc
;; Got recursion not available from 10.96.0.10
Server:                10.96.0.10
Address:                10.96.0.10#53

Name:   wiki-svc.default.svc.cluster.local
Address: 10.108.15.183
;; Got recursion not available from 10.96.0.10

```

The pod test-container is working (representing the test-pod.yaml file). The IP 10.96.0.10 belongs to the ClusterIP range used by Kubernetes Services. This DNS service IP belongs to the Service CIDR.

From the test-container pod, Wiki.js can be reached using its Service name “wiki-svc” or its clusterIP address 10.108.15.183.

```

user@deb:~/Documents/lab2$ kubectl get pods -n kube-system -l k8s-app=kube-dns
NAME                                READY   STATUS    RESTARTS   AGE
coredns-66bc5c9577-mntjn            1/1     Running   2 (5h30m ago)  5h31m

```

CoreDNS pods act as the DNS server for the cluster, as they provide the DNS service in Kubernetes.

```

! wikijs-clusterip.yaml
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: wikijs-clusterip
5  spec:
6    type: ClusterIP
7    selector:
8      app: wikijs
9    ports:
10     - port: 3000
11       targetPort: 3000
12       protocol: TCP
13

```

```

! wikijs-headless.yaml
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: wikijs-headless
5  spec:
6    clusterIP: None #headless
7    selector:
8      app: wikijs
9    ports:
10     - port: 3000
11       targetPort: 3000
12       protocol: TCP
13

```

```

user@deb:~/Documents/lab2$ kubectl apply -f wikijs-clusterip.yaml
service/wikijs-clusterip created
user@deb:~/Documents/lab2$ kubectl apply -f wikijs-headless.yaml
Warning: spec.SessionAffinity is ignored for headless services
service/wikijs-headless created
user@deb:~/Documents/lab2$ kubectl get svc

```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	7h11m
postgres	ClusterIP	10.109.107.105	<none>	5432/TCP	3h16m
wiki-svc	NodePort	10.108.15.183	<none>	3000:30080/TCP	6h6m
wikijs-clusterip	ClusterIP	10.106.38.63	<none>	3000/TCP	93s
wikijs-headless	ClusterIP	None	<none>	3000/TCP	15s

wikijs-clusterip has a single virtual IP (10.106.38.63) and works inside the cluster.

wikijs-headless has no virtual IP (CLUSTER-IP : None) and DNS resolves directly to the individual pod IPs.

Neither clusterip nor headless services can be accessed externally. Only nodeport or loadbalancer services allow external access.

```
user@deb:~/Documents/lab2$ kubectl delete all --all
pod "postgres-5d45787877-cwmg2" deleted
pod "test-container" deleted
pod "wikijs-85c4ccd4cb-jj5mm" deleted
pod "wikijs-85c4ccd4cb-p2jkg" deleted
pod "wikijs-85c4ccd4cb-xmqdx" deleted
service "kubernetes" deleted
service "postgres" deleted
service "wiki-svc" deleted
service "wikijs-clusterip" deleted
service "wikijs-headless" deleted
deployment.apps "postgres" deleted
deployment.apps "wikijs" deleted
replicaset.apps "wikijs-85c4ccd4cb" deleted
user@deb:~/Documents/lab2$ ^C
user@deb:~/Documents/lab2$ minikube stop
👉 Stopping node "minikube" ...
🔴 Powering off "minikube" via SSH ...
🔴 1 node stopped.
user@deb:~/Documents/lab2$ minikube delete
🔥 Deleting "minikube" in docker ...
🔥 Deleting container "minikube" ...
🔥 Removing /home/user/.minikube/machines/minikube ...
💀 Removed all traces of the "minikube" cluster.
```